0.5 Project Scope Sheet · Group 1 · Fernão de Oliveira

Draft for discussion · 12 November 2026 · T. Santos, L. Batista

Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group

Field	Value
Group	Group 1 · Fernão de Oliveira
Members (student number)	106238 · 107304 · 106645
Baseline	B · Azores Feeder: geared container ship, 130 TEU, battery-electric with swappable battery containers, two island loops plus a transit each week
Specialization 1	SP01 · Longitudinal Strength Calculation
Specialization 2	SP02 · Propulsion Matching
Deadline	Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com
ZIP name	Group1_Project.zip
Presentation	Dec 18, 2026, 15 minutes per group

Revisions

Rev.	Date	Description	By
0	To be set	First issue, after discussion with the students	T. Santos, L. Batista

Baseline B

Deliverable Groups D1–D8 and D10

#	Deliverable Group	Description
D1	Design Basis	130 TEU at 10 t per TEU (1,300 t cargo) · battery containers as extra deadweight, mass from supplier data · 11–12 kn · two loops plus the transit in a week · limits at all eight ports, each with its source · every requirement tagged Given / Assumed / To be checked
D2	Reference Ship Database	At least 8 small geared container ships and battery ships · other families as your chosen hull needs · sources for every ship · your own regressions and bounds
D3	Ship Model	Inputs, outputs and calculation sequence (volume carrier) · hull form coefficients (C_B, C_P, C_M, C_WP, vertical prismatic) · tonnage (GT, NT) · lightship weight: steel, outfit, machinery and superstructure, crane included, with centres · deadweight components: cargo, batteries as deadweight, lubricating oil, fresh water, provisions and stores, crew · propulsive coefficients (wake, thrust deduction, relative-rotative and hull efficiency) · propulsive power by Holtrop & Mennen, electric drive, two shafts · propeller selection with a cavitation check · side thruster · electric power estimate for each operating condition · subdivision and compartments (bulkheads, engine room and peaks) · capacities of holds and tanks, containers in the holds and on deck · closed weight balance
D4	Voyage and Financial Model	Both loops and the transit: energy per leg, battery swaps and charging, electricity and port costs · building cost, financing, RFR and NPV · ship model outputs as input
D5	Validation Note	Your model against the reference families you chose, with the choice justified · error per quantity · model corrected where needed
D6	Optimization Report	Minimum battery capacity that completes both loops in one week · justified objective · active constraints · optimum against the requirements · code flowchart: a flowchart of your code that shows how the information flows (inputs, each calculation step, the iterations and the outputs), in the symbols of ISO 5807

#	Deliverable Group	Description
D7	Compliance Sheet	EEDI and CII: show why they do not apply, and report energy per TEU-mile · battery rules · stability while the crane lifts · freeboard · draught at every port
D8	General Arrangement and Hull Form	Deck container layout and crane reach · battery slots aft · container count on your own arrangement · 3D hull form, justified · general arrangement (profile, midship section and main deck), showing the container and battery layout and the crane reach · 3D hull shape with the containers in the holds and on deck, the main cargo-hold bulkheads, the collision bulkhead and the deckhouse exterior · hydrostatics check
D10	Datasheet and Presentation	One-page A4 datasheet (0.4 Datasheet Template) · report on the 0.3 Report Template · oral presentation of the results in PowerPoint, no longer than 15 minutes, including: objectives of the work (description of the initial concept of the ship); computation model (the computation process, with the adopted methods, criteria and validations); assessment of the results obtained (comparison with existing similar ships); conclusions

Evaluation criteria

#	What is evaluated
D1	The transportation problem: ship type and target capacity, the sequence of port calls (round trip), the relevant characteristics of the ports
D2	The ship database compiled and the analysis of its statistical data
D3	Sequence and extent of the computations · selection of the estimation methods · correct response of the outputs to changes in the inputs (qualitative) · every item estimated, with the method named and its range of validity respected
D4	Times and costs at each port · times and energy consumption at sea · global results for one round trip and for one year of operation · financial analysis
D5	Validation of the ship model: lightship weight, propulsive and electric generation power, newbuilding cost
D6	Problem correctly defined: design variables, restrictions (with their sources), objective(s) · exploration of the domain of the variables · the code delivered reproduces the results presented · the flowchart matches the code, shows the inputs, calculation steps, iterations and outputs, and uses the ISO 5807 symbols correctly
D7	Each applicable rule identified with its source, and the result shown against the limit (freeboard, stability with the crane lifting, draught at every port) · EEDI and CII shown not to apply, with the reason
D8	General arrangement with the main dimensions and the cargo area configuration · final main dimensions and hydrostatic properties (displacement, C_B) against the targets, with the error levels discussed · final cargo capacity (TEU, deadweight) against the targets, with the error level discussed
D10	Report: a clear description of all the work, following the 0.3 Report Template · summary of the resulting ship characteristics · comparison with the statistical data of existing ships of the same market segment · presentation as described in D10

Deliverables

ID	Title	Deliverable	Status
STU-D1.1	Owner's requirements and port limits	D1	
STU-D1.2	Laws and regulations applicable, incl. environmental targets	D1	
STU-D2.1	Reference ship database with sources	D2	
STU-D2.2	Regressions and bounds	D2	
STU-D3.1	Ship model (computer codes and/or files)	D3	
STU-D3.2	Weight estimation, assumptions and margins	D3	
STU-D4.1	Voyage model	D4	
STU-D4.2	Financial model	D4	

ID	Title	Deliverable	Status
STU-D5.1	Validation against reference ships	D5	
STU-D6.1	Optimization report against the design objective, sensitivity analysis	D6	
STU-D6.2	Code flowchart (ISO 5807)	D6	
STU-D7.1	EEDI and CII	D7	
STU-D7.2	Preliminary intact stability assessment: requirements and typical conditions, incl. ballast conditions	D7	
STU-D7.3	Draught at every port	D7	
STU-D8.1	Hydrostatics check	D8	
DWG-D8.1	General arrangement (A3): profile, midship section and main deck, with the container and battery layout and the crane reach	D8	
STU-D10.1	Datasheet (template 0.4)	D10	
STU-D10.2	Presentation, 15 minutes	D10	

Specialization scope (individual work)

One calculation note per specialization goes in D9, with the drawings listed. Each specialization has its own self-study deck.

Deliverable Group D9

#	Deliverable Group	Description	What is evaluated
D9	Specialization Documents (2)	One individual report per specialization, with its required studies and drawings (below)	Every required study and drawing delivered · methods and assumptions stated · results checked against the acceptance criteria of the specialization

SP01 · Longitudinal Strength Calculation

Show that the hull girder carries the still-water and wave bending moments.

- STU-SP01.1 Lightship weight and deadweight distribution along the length.
- STU-SP01.2 Four critical loading conditions (subject to justification and/or definition with the instructor).
- STU-SP01.3 Still-water and wave bending moments and shear forces for each loading condition.
- STU-SP01.4 Allowable stresses, in bending and in shear.
- STU-SP01.5 Preliminary scantlings of the midship cross-section.
- STU-SP01.6 Section modulus, moment of inertia and neutral axis of the midship section.
- STU-SP01.7 Allowable vertical still-water bending moment and shear force: the allowable envelopes along the length, with each loading condition plotted against them.
- DWG-SP01.1 Midship cross-section.

ID	SP01 deliverable	Status
STU-SP01.1	Lightship and deadweight distribution	
STU-SP01.2	Four critical loading conditions	
STU-SP01.3	Still-water and wave bending moments and shear forces	
STU-SP01.4	Allowable stresses, bending and shear	
STU-SP01.5	Preliminary scantlings of the midship cross-section	
STU-SP01.6	Section modulus, moment of inertia and neutral axis	
STU-SP01.7	Allowable still-water bending moment and shear force vs loading conditions	
DWG-SP01.1	Midship cross-section	

SP02 · Propulsion Matching

Choose the propeller and match it to the motors in every operating mode.

The SP02 report is a resistance and propulsion report, justifying:

- STU-SP02.1 Detailed ship resistance: each resistance component, the margins and the assumptions, computed from 4 to 14 kn in steps of 1 kn; comparison with the Baseline results.
- STU-SP02.2 Statement of the design, machinery and sea margins.
- STU-SP02.3 Optimum Wageningen B-series propeller for the trial condition, the service speed, the top speed and the mainland transit (7 kn).
- STU-SP02.4 Selection of the propeller design condition, with justification; comparison with the Baseline results.
- STU-SP02.5 Propeller description and features.

- STU-SP02.6 Generator sizing: one generator able to run both electric motors at the mainland transit speed, with and without the sea margin.
- STU-SP02.7 Engine matching: operating points on the motor layout diagram, with the sea margin.
- STU-SP02.8 The two-shaft electric modes (Baseline B).
- DWG-SP02.1 Stern arrangement: propeller, shaft line and rudder.

ID	SP02 deliverable	Status
STU-SP02.1	Detailed resistance, 4 to 14 kn	
STU-SP02.2	Design, machinery and sea margins	
STU-SP02.3	Optimum B-series propeller, four conditions	
STU-SP02.4	Propeller design condition	
STU-SP02.5	Propeller description and features	
STU-SP02.6	Generator sizing for the mainland transit	
STU-SP02.7	Engine matching	
STU-SP02.8	Two-shaft electric modes	
DWG-SP02.1	Stern arrangement	

The full text of every item is in the Statement of Work.